① Aspect 9 (Highest Relevance)

Translate a problem into a standard mathematical representation or algorithm

In *faulty-players-qs3*, students are typically required to:

- Convert descriptions involving **daily production, fault rates, comparisons, or targets**
- Into **explicit mathematical relationships**, such as expressions, calculation procedures, or repeatable algorithms

Unlike qs1, which focuses on understanding given data, qs3 requires students to:

- Actively **rewrite the situation in mathematical form**
- Make explicit *how* the problem should be calculated, not just *what* should be calculated

This makes Aspect 9 the most central *Formulate* competency assessed.

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② Aspect 1 / Aspect 3

Select a mathematical description or an appropriate representation

Students must decide, for example:

- Whether to model video and audio players separately
- Whether to represent faults using percentages, ratios, or absolute quantities
- Which mathematical representation best supports solving the problem

This reflects the ability to **actively choose appropriate mathematical representations**, rather than passively using a provided table.

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③ Aspect 2 / Aspect 6

Identify key variables and mathematical aspects of the situation

In qs3, students must clearly identify and work with key variables such as:

- Daily production quantities for each type of player
- Fault rates for each product type
- Target quantities (e.g. number of faulty units, comparison criteria, or decision thresholds)

Here, identifying variables is not merely about comprehension, but about **directly supporting model construction**.

④ Aspect 7

Simplify or decompose the situation

To create a workable model, students typically:

- Decompose total production into separate video and audio components
- Work through the situation step by step (e.g. calculate faulty units first, then compare or aggregate)

This demonstrates the *Formulate* skill of **breaking a complex situation into manageable sub-problems**.

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⑤ Aspect 12

Identify assumptions, constraints, and simplifications

At a higher level of response, students may recognise that:

- The model uses averages rather than exact daily values
- Video and audio player production are treated as independent
- Variability in production or testing accuracy is ignored

This reflects awareness of the **conditions under which the model is valid and its limitations**, a higher-order *Formulate* competency.

① Aspect 9（相关性最高）

Translate a problem into a standard mathematical representation or algorithm

📌 在 faulty-players-qs3 中，学生通常需要：

- 将关于“日产量、故障率、比较或目标”的描述
- 转化为**明确的数学关系**（如表达式、计算步骤或可重复的算法）

👉 与 qs1 不同，这里不再只是“理解表格”，而是：

- 主动把问题**转写为数学形式**
- 明确“如何算”，而不仅是“算什么”

这使 Aspect 9 成为 qs3 中最核心的 formulate 能力。

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② Aspect 1 / Aspect 3

Select a mathematical description or an appropriate representation

📌 学生需要决定：

- 是否分别建模 video 与 audio players
- 是否用百分比、比值或绝对数量来表示故障情况
- 哪种数学表示最有助于解决问题

👉 这体现了在给定情境中 **主动选择合适数学表示** 的能力，而非被动使用已给出的表格。

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③ Aspect 2 / Aspect 6

Identify key variables and mathematical aspects of the situation

📌 在 qs3 中，学生必须清楚识别并使用关键变量，例如：

- 每种播放器的日产量
- 各自的故障率
- 目标量（如故障数、比较结果或决策标准）

👉 变量识别不再只是“看懂”，而是直接服务于模型构建。

④ Aspect 7

Simplify or decompose the situation

📌 为了形成可操作的模型，学生通常需要：

- 将“整体生产”拆分为 video / audio 两个子系统
- 分步骤处理（先算故障数，再比较或汇总）

👉 这是典型的 **通过分解来实现数学建模** 的 formulate 行为。

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⑤ Aspect 12

Identify assumptions, constraints, and simplifications

📌 在更高水平的回答中，学生可能意识到：

- 使用的是平均值而非每日真实数据
- 假设两类产品相互独立
- 忽略了生产波动或检测误差

👉 这反映了对模型**适用条件与局限性**的认识，是 formulate 的高阶表现。